

Cluster-Guided Deep Learning for Baseflow Prediction in Indian River Basins Using CAMELS-IND

1. Project Overview

Problem Statement

Baseflow -- the portion of streamflow sustained by groundwater discharge -- is a critical hydrological variable for water resource management, drought assessment, and ecological flow estimation. Predicting baseflow at ungauged or sparsely gauged catchments across India remains a significant challenge due to the heterogeneous hydroclimatic and physiographic landscape of the subcontinent.

Motivation

India's river systems span diverse climatic zones, geological formations, and land-use patterns. Traditional lumped hydrological models often fail to generalize across such heterogeneity. This project investigates whether **clustering catchments by static attributes** and subsequently training **cluster-specific deep learning models** can yield more accurate and interpretable baseflow predictions compared to a single global model.

Real-World Relevance

- Water resource planning for 242 catchments across India
- Baseflow estimation supports irrigation scheduling, reservoir operations, and environmental flow requirements
- Regional clustering enables transfer learning to ungauged basins within the same hydroclimatic cluster

Task Type

- **Primary task:** Time-series regression (daily baseflow prediction)
- **Supporting task:** Unsupervised clustering of catchments by static attributes

2. Dataset

Source

The project uses the **CAMELS-IND** (Catchment Attributes and MEteorology for Large-sample Studies -- India) dataset, a large-sample hydrology dataset analogous to CAMELS-US, CAMELS-GB, and CAMELS-CL.

Data Composition

Static Attributes (211 features per catchment):

Category	Example Features	Count
Topography	elev_mean, slope_mean, dpsbar	~5
Climate	p_mean, aridity_p_pet, high_prec_freq, low_prec_freq	~12
Hydrology	bfi, runoff_ratio, q_mean, mean_anum_flow, freq_q_low, freq_q_high	~15
Soil	soil_depth, soil_awc_top, soil_awc_sub, sand_frac_top, clay_frac_top, bulkdense_top_mean	~18
Geology	geol_porosity, geol_permeability, geol_class_1st, geol_class_2nd	~5

Category	Example Features	Count
Land Cover	crops_frac, trees_frac, water_frac, built_area_frac, lai_mean	~10
Anthropogenic	num_dams, reservoir_index, irrigation_frac, pop_density_2020	~5

Dynamic Forcing Features (18 time-varying features, daily resolution):

Feature	Unit
prcp	mm/day
tmax, tmin, tavg	Celsius
srad_lw, srad_sw	W/m ²
wind_u, wind_v, wind	m/s
rel_hum	%
pet_gleam, aet_gleam	mm/day
evap_canopy, evap_surface	mm/day
sm_lvl1 through sm_lvl4	kg/m ²

Target Variable: Baseflow (mm/day), extracted via the Boughton baseflow separation method applied to observed streamflow records.

Dataset Statistics

Property	Value
Total catchments	242 (filtered for >30% flow data availability)
Temporal coverage	1980-01-01 onwards (daily)
Static feature dimensions	211 (198 numerical, 13 categorical)
Dynamic feature dimensions	18
Missing values in static data	3,463 (imputed with column median)
Total rows across all clusters	2,875,392
Columns per combined cluster file	74

Clustering-Based Data Splits

After clustering (see Section 3), per-cluster data volumes (from output/merged_clustered/cluster_summary_new.csv, the merged dataset actually used for training) are:

Cluster	Gauges	Files Found	Rows	Dominant Basins
C0	65	56	838,656	Krishna, Cauvery, EFRS
C1	66	49	733,824	Godavari, Mahanadi, Brahmani-Baitarani
C2	37	36	539,136	WFRS, Cauvery, Krishna
C3	74	51	763,776	Northern basins
Total	242	192	2,875,392	

Note: "Gauges" is the count of catchments assigned to each cluster; "Files Found" is the number of per-gauge dynamic CSVs successfully located and merged. Not every assigned gauge has a usable dynamic record, which is why files < gauges in some clusters and the totals differ.

Data Preprocessing Pipeline

1. **Static data assembly:** Merged 211 attributes from CAMELS-IND attribute CSV files for all 242 gauges
2. **Dynamic data merging:** Combined daily forcing data with baseflow time series per gauge, joined on date
3. **Missing value handling:** Static features imputed with median; dynamic NaN rows filled with column mean
4. **Categorical encoding:** 11 categorical static features (e.g., geol_class_1st, dom_land_cover, river_basin) encoded via OrdinalEncoder
5. **Normalization:** StandardScaler applied to dynamic features; GroupNormalizer with softplus transformation for TFT target
6. **Temporal indexing:** Created time_idx for sequential ordering; extracted month, day, year as known categorical covariates

Challenges

- **Missing data:** 3,463 missing static attribute values across 242 gauges
- **Heterogeneous record lengths:** Not all gauges have continuous daily records from 1980; filtered to gauges with >30% data availability
- **Extreme values:** Baseflow ranges span several orders of magnitude (e.g., top values exceed 19,000 mm/day for large basins)

3. Methodology

Overall Approach

The project follows a **cluster-then-predict** paradigm:

1. **Dimensionality reduction** of 197 numerical static attributes using UMAP (to 40 dimensions)
2. **Unsupervised clustering** of 242 catchments into hydroclimatically homogeneous groups
3. **Cluster-specific model training** using Temporal Fusion Transformers (TFT) and LSTMs
4. **Cross-validated evaluation** with expanding-window time-series splits

Clustering Pipeline

Dimensionality Reduction: - PCA analysis showed 52 components needed for 95% variance; top 40 components explain 91.7% - UMAP tested at 2, 5, 10, 20, 40 dimensions; 40D selected for best distance correlation (0.4675) - UMAP configuration: n_neighbors=15, min_dist=0.1, random_state=42

Clustering Methods Compared:

Method	Description
K-Means	Centroid-based partitioning with silhouette-optimized k
Gaussian Mixture Model (GMM)	Probabilistic soft clustering with full covariance
Enhanced Growing Neural Gas (GNG)	Topology-preserving unsupervised neural network
HDBSCAN	Density-based hierarchical clustering (min_cluster_size=10, min_samples=5)
DBSCAN	Density-based clustering tested across multiple eps values (0.1--3.0)

Silhouette analysis over k=2 to k=19 identified k=4 as optimal. K-Means with k=4 was selected for producing geographically coherent regions aligned with known river basin boundaries.

Geographic Distribution of Clusters:

Cluster	Lat Range	Lon Range	Center	Dominant Basins
C0	9.32 - 20.20	72.93 - 80.82	(14.70, 77.05)	Krishna, Cauvery, EFRS
C1	16.79 - 23.20	73.65 - 86.92	--	Godavari, Mahanadi, Brahmani-Baitarani
C2	8.31 - 15.47	74.10 - 77.83	--	WFRS, Cauvery, Krishna
C3	Northernmost gauges	--	--	Northern basins

Model Architectures

3.1 LSTM Baseline (Single-Gauge)

A 3-layer stacked LSTM trained on gauge 03005 as a proof-of-concept:

```
Input (seq_len=20, features=220)
-> LSTM(220, 256) -> Dropout(0.5) -> ReLU
-> LSTM(256, 128) -> Dropout(0.5) -> ReLU
-> LSTM(128, 32) -> Dropout(0.5) -> ReLU
-> Linear(32, 1)
```

The 220 input features comprise 27 dynamic features + 193 static features (replicated across the temporal dimension and concatenated).

3.2 Temporal Fusion Transformer (TFT) -- Primary Model

The TFT is an attention-based architecture designed for multi-horizon time-series forecasting with built-in interpretability. Key components include:

- **Variable Selection Networks:** Learns which input features are most relevant at each time step
- **Gated Residual Networks (GRN):** Provides nonlinear processing with skip connections
- **Multi-Head Attention:** Captures long-range temporal dependencies across the encoder window
- **Static Enrichment:** Integrates static catchment attributes into temporal processing

TFT Hyperparameters:

Hyperparameter	Value
Hidden size	108
Attention heads	2
Dropout	0.2
Hidden continuous size	8
Encoder length	18 (also tested 14, 10)
Prediction length	1
Learning rate	0.001
Parameters	~843.7k
Loss function	RMSE
Target normalization	GroupNormalizer (softplus)

Feature configuration in TFT (as set in training/cv/tft-cv-model.ipynb): - **Static reals:** 39 catchment attributes — the *without-hydrological-signature* feature set is the one active in the notebook (elevation, slope, ghi_area, dpsbar, precipitation climatology, soil texture/water-holding, geology, land cover, wtd, total_storage, reservoir_index). The 49-feature variant that additionally includes the hydrological summaries (bfi, runoff_ratio, q_mean, flow statistics) is present but commented out. - **Static categoricals:** none (static_categoricals=[]) - **Time-varying known reals:** 18 dynamic forcing variables - **Time-varying known categoricals:** month, day, year (cast to string) - **Time-varying unknown reals:** baseflow (target) - **Target normalizer:** GroupNormalizer(groups=["group"], transformation="softplus"), with add_relative_time_idx, add_target_scales, and add_encoder_length enabled

3.3 KNN-Guided Models

Hybrid approaches combining K-nearest-neighbor similarity search with deep learning:

- **KNN + TFT:** Identifies similar gauges via KNN, trains TFT on combined data
- **KNN + Combined LSTM:** KNN-selected gauge data fed into shared LSTM
- **KNN + Front LSTM:** KNN-based pre-filtering with LSTM prediction head

Key Design Decisions

1. **UMAP over PCA:** PCA requires 52 components for 95% variance; UMAP at 40 dimensions achieves superior distance correlation (0.4675) while preserving nonlinear manifold structure
2. **K-Means over alternatives:** K-Means with k=4 chosen for interpretable, geographically coherent clusters with zero noise assignment
3. **Per-cluster TFT models:** Training separate models per cluster reduces intra-cluster heterogeneity and improves model capacity allocation
4. **Expanding-window CV:** Respects temporal ordering; avoids future data leakage inherent in random splits
5. **Feature ablation (w/o hydro):** Tested TFT without hydrological static features (bfi, runoff_ratio, q_mean,

4. Training Pipeline

Data Loading

- Per-cluster combined CSV files (cluster_{cluster}_combined_new.csv, 74 columns each) loaded from output/merged_clustered/
- NaNs filled with column means (df.fillna(df.mean(numeric_only=True))); cluster and gauge_id columns dropped; month/day/year cast to string
- Data split temporally: 70% train / 15% validation / 15% test (the active configuration). A 5-fold expanding-window CV scaffold also exists but is commented out

Transformations

- StandardScaler on all dynamic features (LSTM pipeline)
- GroupNormalizer with softplus transformation on baseflow target (TFT pipeline)
- Categorical features (month, day, year) provided as known future covariates

Loss Functions

Loss	Formula	Used By
LogRMSE	$\sqrt{\text{mean}((\log(\text{clamp}(\text{pred})) - \log(\text{clamp}(\text{target})))^2)}$	- LSTM baseline
RMSE	$\sqrt{\text{mean}((\text{pred} - \text{target})^2)}$	TFT primary model

LogRMSE was designed specifically for this project to handle the orders-of-magnitude variation in baseflow values; predictions and targets are clamped to a minimum (eps=1e-8 in the TFT-notebook implementation) before log transformation. A custom LogRMSE Metric subclass is defined in the TFT notebook but the model is trained with plain RMSE(); LogRMSE serves as the LSTM-baseline loss. **KGE is computed only as a post-hoc diagnostic metric (via kge_loss), not as a training objective.**

Training Configuration

LSTM (Baseline):

Setting	Value
Optimizer	Adam
Learning rate	1e-4
Batch size	32
Epochs	1000
Loss function	LogRMSE
Sequence length	20
Device	CUDA GPU

TFT (Primary):

Setting	Value
Framework	pytorch_forecasting + PyTorch Lightning (lightning.pytorch)
Optimizer	Adam (per saved model card)
Learning rate	0.001
LR scheduling	ReduceOnPlateau (reduce_on_plateau_patience=4)
Gradient clipping	gradient_clip_val=0.1
Early stopping	Monitored on val_loss (patience 10, min_delta=1e-4)
Max epochs	20 (model card)
Batch size	1024 (train); validation/test/inference at batch_size × 10
Device	CUDA GPU (Kaggle)
Seed	42 (pl.seed_everything(42))

Data-Splitting Strategy

Correction vs. earlier draft: The training notebook (tft-cv-model.ipynb) runs a **single temporal split**, not an automated multi-fold loop. The active TimeSeriesDataSet setup uses a chronological **70% / 15% / 15%** train / validation / test split (train_cutoff = $0.7 \cdot N$, val_cutoff = $0.85 \cdot N$). The best checkpoint (by validation loss) is then used to predict over the full cluster series, and final RMSE/MAE/R²/KGE are computed on that.

An expanding-window 5-fold cross-validation scaffold is also present in the notebook (fold setup in the CV-configuration cell), designed as follows:

Fold	Training Window	Validation Window
1	0 -- 50% of data	50% -- 65%
2	0 -- 57%	57% -- 72%
3	0 -- 64%	64% -- 79%
4	0 -- 71%	71% -- 86%
5	0 -- 78%	78% -- 93%

The fold split logic is $\text{train_end} = 0.5 \cdot N + (\text{fold}/5) \cdot (0.35 \cdot N)$, $\text{val_end} = \text{train_end} + 0.15 \cdot N$, so training data grows each fold from a 50% minimum and validation windows are 15% slices. **However, the fold-training loop itself is commented out in the current notebook**, so the reported per-cluster metrics come from the single-split best checkpoint rather than an averaged 5-fold run. The model artifacts are nonetheless named ...-cv5fold-seq18-... reflecting the intended CV protocol.

Experiment Tracking

- **Platform:** Weights & Biases (WandB)
- **Projects:** spark'25-lstm-testing (LSTM), spark'25-lstm-random (TFT)
- **TFT run naming:** c-{cluster}-tft-cv-{run_num}-seq-18-w/o-hydro
- **Logged:** train_loss, val_loss per epoch; final test metrics (test_rmse, test_mae, test_r2, test_kge); interactive prediction plots; predicted-vs-actual scatter; attention/variable-importance interpretation figures

Hardware

- Training executed on **Kaggle GPU instances** (CUDA-enabled)
- Local development and preprocessing on Windows 11 with Anaconda/Python environment

5. Experiments

Experiment 1: LSTM Baseline on Gauge 03005

Property	Detail
Hypothesis	A single-gauge LSTM with concatenated static features can learn baseflow dynamics
Configuration	3-layer LSTM, 220 features, seq_len=20, LogRMSE loss
Training data	14,555 daily records (80/10/10 split) -> 11,644 train / 1,456 val / 1,389 test sequences

Experiment 2: Clustering Analysis

Property	Detail
Hypothesis	Indian catchments can be meaningfully grouped by static attributes for regionalized modeling
Methods compared	K-Means, Enhanced GNG, GMM, HDBSCAN, DBSCAN

Property	Detail
Dimensionality reduction	UMAP: 197 features -> 40 dimensions (distance correlation: 0.4675)
Optimal clusters	k=4 via silhouette analysis

Experiment 3: TFT with 5-Fold Cross-Validation (Per Cluster)

Property	Detail
Hypothesis	TFT with expanding-window CV provides robust baseflow predictions within a hydroclimatic cluster
Configuration	hidden_size=108, attention_heads=2, dropout=0.2, hidden_continuous=8, encoder_len=18, pred_len=1, loss=RMSE, batch=1024
Clusters trained	C0 (838,656 rows), C1 (733,824 rows), C2 (539,136 rows), C3 (763,776 rows)
Split	Single chronological 70/15/15 split; best-by-val_loss checkpoint used for full-series prediction (5-fold CV scaffold present but commented out)

Experiment 4: TFT without Hydrological Features (Ablation)

Property	Detail
Hypothesis	Removing hydrological static features (bfi, runoff_ratio, q_mean, etc.) tests whether the model relies on potentially circular information
Removed features	freq_q_low, freq_q_high, mean_anum_flow, bfi, runoff_ratio, q_mean, mean_swmn_flow, mean_atmn_flow, mean_wint_flow, mean_sumr_flow
Remaining static features	39 (vs. 49 with hydro)
Tested configurations	seq_len=18, seq_len=14, seq_len=10
Note	The 39-feature without-hydro set is the configuration used for the final exported models (run names tagged w/o-hydro); the 49-feature with-hydro list is retained in the notebook (commented out) as the comparison baseline

Experiment 5: KNN + TFT Combined Model

Property	Detail
Hypothesis	Nearest-neighbor pre-selection of similar gauges improves prediction
Configuration	KNN-based gauge selection + TFT training

Experiment 6: KNN + LSTM Variants

Property	Detail
Hypothesis	KNN-guided LSTM models can leverage inter-gauge similarity for improved predictions
Variants	KNN + Combined LSTM (shared LSTM across KNN-selected gauges), KNN + Front LSTM (KNN pre-filtering with LSTM head)

6. Evaluation Metrics

Metrics Used

Metric	Formula	Rationale
RMSE	$\sqrt{\text{mean}((y - y_{\text{hat}})^2)}$	Primary loss; penalizes large errors heavily -- important for flood/baseflow extremes
MAE	$\text{mean}(y - y_{\text{hat}})$	Robust to outliers; interpretable in original units (mm/day)
R ²	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$	Proportion of variance explained; benchmark against mean prediction
KGE	$1 - \sqrt{((r-1)^2 + (\alpha-1)^2 + (\beta-1)^2)}$	Decomposes into correlation (r), variability ratio (alpha), bias ratio (beta); standard in hydrology
NSE	$1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$	Equivalent to R ² for time series; widely used in hydrology
LogRMSE	$\sqrt{\text{mean}((\log(y) - \log(y_{\text{hat}}))^2)}$	Handles orders-of-magnitude variation in baseflow; used for LSTM baseline

Metric Selection Justification

- **KGE** is the preferred metric in modern hydrology (Gupta et al., 2009) as it addresses known deficiencies of NSE (sensitivity to bias and variability)
- **RMSE** used as the training loss for TFT due to direct gradient properties
- **LogRMSE** used for LSTM to handle the highly skewed baseflow distribution

7. Results & Analysis

Per-Cluster TFT Performance (seq_len=18, w/o hydrological features)

The final per-cluster TFT models (encoder length 18, RMSE loss, without-hydro static set) were exported to Hugging Face; the full-series RMSE is encoded in each artifact name and recorded in the prediction script (DEFAULT_MODELS):

Cluster	Model Artifact	Final RMSE (mm/day)	Rows
C0	tft-baseflow-c0-cv5fold-seq 18-rmse185.0666	185.07	838,656
C1	tft-baseflow-c1-cv5fold-seq 18-rmse665.7481	665.75	733,824
C2	tft-baseflow-c2-cv5fold-seq 18-rmse317.5040	317.50	539,136
C3	tft-baseflow-c3-cv5fold-seq 18-rmse90.7000	90.70	763,776

Observations:

- **C3 is the best-performing cluster** (RMSE 90.70), and **C1 the worst** (RMSE 665.75). RMSE here is in the target's native units and is dominated by the large-magnitude baseflow values in each cluster (recall the largest baseflow values exceed several thousand mm/day), so absolute RMSE is not directly comparable across clusters with different flow-magnitude distributions.
- The wide spread (90 → 666) reflects intra-cluster heterogeneity in flow scale rather than purely model quality; per-cluster R²/KGE (logged to WandB as test_r2/test_kge) are the more comparable cross-cluster indicators and should be filled in from the run dashboard.

RMSE/MAE/R²/KGE for the held-out 15% test split and the full-series evaluation are logged per run to WandB (test_rmse, test_mae, test_r2, test_kge) and saved in each model's model_card.json on Hugging Face. Populate the table below from those sources to complete the analysis.

Cluster	Test RMSE	Test MAE	Test R ²	Test KGE
C0	(fill)	(fill)	(fill)	(fill)
C1	(fill)	(fill)	(fill)	(fill)
C2	(fill)	(fill)	(fill)	(fill)
C3	(fill)	(fill)	(fill)	(fill)

8. Ablation Studies

Ablation 1: Hydrological Feature Removal

- **What was changed:** Removed 10 hydrological static features (bfi, runoff_ratio, q_mean, freq_q_low, freq_q_high, mean_anum_flow, mean_swmn_flow, mean_atmn_flow, mean_wint_flow, mean_sumr_flow)
- **Why:** These features are derived from the streamflow signal being predicted, introducing potential circularity. Testing without them assesses whether the model can predict baseflow from purely exogenous catchment attributes.
- **Configurations tested:** With 49 static features (full) vs. 39 static features (w/o hydro)

Ablation 2: Encoder Sequence Length

- **What was changed:** Encoder context window varied across 18, 14, and 10 time steps
- **Why:** Longer sequences capture antecedent moisture conditions and seasonal patterns but increase computational cost and may introduce noise
- **Configurations tested:** seq_len = {18, 14, 10}, with and without hydrological features

Ablation 3: Clustering Method

- **What was changed:** Compared K-Means, GMM, GNG, HDBSCAN, and DBSCAN for catchment grouping
- **Why:** Different clustering algorithms make different assumptions about cluster geometry (convex vs. arbitrary shape, fixed vs. variable number of clusters)
- **Outcome:** K-Means selected for producing geographically coherent 4-cluster partition

Quantitative ablation results to be populated.

9. Limitations

Model Limitations

- **Single-step prediction:** The model predicts only 1 day ahead; multi-horizon forecasting was not explored
- **No ensemble methods:** Each cluster uses a single best-fold model; ensembling across folds could improve robustness
- **Fixed cluster assignment:** Catchments near cluster boundaries may be misclassified; soft clustering or overlapping approaches were not used

Dataset Limitations

- **Baseflow separation method:** Baseflow is not directly observed but estimated via the Boughton method; errors in separation propagate to model training
- **Temporal coverage gaps:** Not all 242 gauges have continuous records; missing periods are filled rather than masked
- **Static feature circularity:** Including bfi, runoff_ratio, and q_mean as static features is potentially circular since they are derived from the streamflow signal being predicted
- **Single dataset:** No external validation on independent catchments outside CAMELS-IND

Generalization Issues

- Models are trained and evaluated within the CAMELS-IND domain; transferability to other regions (e.g., CAMELS-US, CAMELS-GB) is unknown
 - Cluster assignments are fixed; catchments near cluster boundaries may be misclassified
-

10. Future Work

1. **Multi-horizon prediction:** Extend prediction length beyond 1 day to support operational forecasting (7-day, 30-day horizons)
 2. **Sub-cluster refinement:** For underperforming clusters, introduce hierarchical sub-clustering or gauge-weighted loss functions
 3. **Ensemble across CV folds:** Average predictions from all 5 folds to reduce variance
 4. **Transfer learning:** Pre-train on all clusters, fine-tune on target cluster to leverage shared hydrological knowledge
 5. **Alternative architectures:** Evaluate Transformers (PatchTST, iTransformer), Neural ODE-based models, and graph neural networks that encode spatial catchment connectivity
 6. **Remove circular features:** Train production models exclusively without hydrological summary statistics
 7. **Uncertainty quantification:** Leverage TFT's quantile regression capability (currently unused) for probabilistic predictions
 8. **Scaling:** Extend to all CAMELS-IND catchments (not just 242 with >30% availability) using data imputation or semi-supervised approaches
 9. **Real-time deployment:** Package the best models as a prediction service for operational hydrology agencies (CWC)
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11. Reproducibility

Steps to Reproduce

Data preparation: - Obtain CAMELS-IND dataset (static attributes + dynamic forcings + streamflow) - Run `data_preprocess.ipynb` to merge static and dynamic features per gauge - Output: `gauge_data/static_all_gauges.csv`, per-gauge dynamic CSVs

Clustering: - Run `clustering.ipynb` - Produces: `output/clustered_gauges.csv` (242 gauges with cluster labels) - Produces: `output/merged_clustered/cluster_*_combined_new.csv`

Model training: - Run `training/cv/tft-cv-model.ipynb` on Kaggle (or any CUDA-enabled environment) - Set `cluster = {0, 1, 2, 3}` (and `run_num`) at the top of the notebook to train per cluster - The notebook loads `cluster_{cluster}_combined_new.csv`, fills NaNs with column means, builds the `TimeSeriesDataSet` on a single 70/15/15 chronological split, trains the TFT, and uploads the best checkpoint to Hugging Face - (Optional) Uncomment the CV-loop cell to run the 5-fold expanding-window protocol instead of the single split

Inference / Evaluation: - Use `predict.py` (repo root; the standalone version of the notebook's prediction system). It downloads the per-cluster model + `model_card.json` from Hugging Face (DEFAULT_MODELS), rebuilds the `TimeSeriesDataSet`, runs predictions, prints RMSE/MAE/R²/KGE, and writes a `y_true/y_pred` CSV - Per-cluster predictions saved as `seq18-predictions_c{cluster}_complete.csv` - Attention / variable-importance interpretations logged to WandB and saved under `plots_results/`

Dependencies

```
pandas
numpy
scikit-learn
torch (PyTorch)
pytorch-forecasting
pytorch-lightning (lightning)
umap-learn
hdbscan
matplotlib
seaborn
plotly
folium
geopandas
shapely
```

Configuration

Parameter	Value
Random seed	42
TFT hidden size	108
TFT attention heads	2
TFT dropout	0.2
TFT hidden continuous size	8
Encoder length	18
Prediction length	1
Batch size	1024
Train/val/test split	0.70 / 0.15 / 0.15 (chronological)
CV folds (scaffold)	5 (expanding window, min train fraction 0.5) — currently disabled
Max epochs	20
Static features used	39 (without-hydro set)
UMAP components	40
K-Means clusters	4

Model Artifacts

Trained per-cluster TFT models uploaded to Hugging Face under user akshatshaw, named tft-baseflow-c{cluster}-cv5fold-seq18-rmse{RMSE}-timestamp{ts}:

- C0 — tft-baseflow-c0-cv5fold-seq18-rmse185.0666-timestamp20260328_113728
- C1 — tft-baseflow-c1-cv5fold-seq18-rmse665.7481-timestamp20260329_111459
- C2 — tft-baseflow-c2-cv5fold-seq18-rmse317.5040-timestamp20260330_074905
- C3 — tft-baseflow-c3-cv5fold-seq18-rmse90.7000-timestamp20260331_073611

Each repo contains model.pth (state dict) and model_card.json (architecture, feature lists, metrics).

Experiment Tracking

All runs logged to WandB: - Project: spark'25-lstm-testing (LSTM experiments) - Project: spark'25-lstm-random (TFT experiments) - User: akshatshaw47-iit-roorkee